WEEK 1 DLP EXERCISE

✅ (a) AI vs ML vs DL — Real-Life Examples

i) Artificial Intelligence (AI)

Example: A rule-based chess engine

- Uses predefined rules + search
- Doesn't necessarily learn from data
- Goal = intelligent behavior

👉 AI = broad field of making machines act intelligently

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ii) Machine Learning (ML)

Example: Email spam filter trained on labeled emails

- Learns patterns from data
- Improves with more examples

👉 ML = AI that learns from data

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iii) Deep Learning (DL)

Example: Face recognition in phone unlock

- Uses deep neural networks
- Learns features automatically from images

👉 DL = ML using deep neural networks

✅ (b) Linear vs Non-Linear Functions

Linear = straight line form → $ax + b$
Non-linear = curves, powers, logs, etc.

i) **y = 3x + 2** → ✅ Linear
ii) **y = x^2 + 1** → ❌ Non-linear (power 2)
iii) **y = log(x)** → ❌ Non-linear (log curve)
iv) **y = 5** → ✅ Linear (constant = slope 0 line)

✅ (c) Supervised vs Unsupervised

Supervised = labeled output exists
Unsupervised = no labels, only patterns

i) Predict house price → ✅ Supervised (price label known)
ii) Group students by habits → ✅ Unsupervised (clustering)
iii) Spam detection → ✅ Supervised (spam/not spam labels)
iv) Customer segments → ✅ Unsupervised (pattern grouping)

✅ (d) Manhattan Distance

Formula:
|x1-x2| + |y1-y2|

P(1,2), Q(6,5)

= |1−6| + |2−5|
= 5 + 3
= **8**

✅ (e) Euclidean Distance

Formula:
sqrt{(x1-x2)^2 + (y1-y2)^2}

= $\sqrt{[(1-6)^2 + (2-5)^2]}$
= $\sqrt{(25 + 9)}$
= $\sqrt{34}$
= **5.83**

✅ (f) Confusion Matrix Metrics

Given:
TP=12, FP=3, TN=30, FN=5
Total = 50

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i) Accuracy

[
(TP+TN)/Total = (12+30)/50 = 42/50 = 0.84
]
✅ **84%**

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ii) Precision

[
TP/(TP+FP) = 12/(12+3) = 12/15 = 0.80
]
✅ **80%**

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iii) Recall

[
TP/(TP+FN) = 12/(12+5) = 12/17 = 0.706
]
✅ **70.6%**

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iv) F1 Score

[
2PR/(P+R)
]

= 2×0.80×0.706 / (1.506)
= **0.75**

✅ (g) Cancer Detection — Precision vs Recall

Correct: **Recall**

Because:

- Recall reduces **False Negatives**
- Missing a cancer case is dangerous
- False alarms are safer than missed disease

👉 Choose **Recall**

✅ (h) Threshold Increase (0.3 → 0.9)

Higher threshold = model becomes stricter before saying "positive"

i) False Positives → ⬇ Decrease
ii) False Negatives → ⬆ Increase
iii) Precision → ⬆ Increase
iv) Recall → ⬇ Decrease

Memory trick:

High threshold = fewer YES predictions

✅ (i) Class Imbalance

Data:
990 negative
10 positive
Model predicts all negative

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i) Accuracy

Correct predictions = 990
Total = 1000

Accuracy = 990/1000 = **99%**

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ii) Why useless?

- Model **never detects positive class**
- Completely fails the important minority class
- Looks good only because data is imbalanced

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iii) Better Metrics

Use:

- ✅ Recall
- ✅ Precision
- ✅ F1 Score
- ✅ Confusion Matrix
- ✅ ROC-AUC / PR-AUC

Because these show performance on minority class.

We'll use your same imbalanced dataset example:

- Total = 1000 samples
- Positive (minority) = 10
- Negative (majority) = 990
- Model predicts **all negative**

Confusion matrix:

- TP = 0
- FP = 0
- TN = 990
- FN = 10

Accuracy = **99%** ❌ (looks great but model is useless)

Now let's see how each "better metric" exposes the failure.

✅ Why These Metrics Reveal Minority-Class Failure

🎯 1 ⬜ Recall (Sensitivity)

Formula:
[
Recall = TP / (TP + FN)
]

Measures:

Out of all actual positives — how many did we catch?

Here:
TP = 0, FN = 10

Recall = 0 / 10 = **0**

✅ Instantly shows:

Model detected **ZERO** minority cases

That's why recall is powerful for minority class — it looks directly at positive-class coverage.

🎯 2 Precision

Formula:
[
Precision = TP / (TP + FP)
]

Measures:

When model predicts positive — how often is it correct?

Here:
Model never predicts positive → TP+FP = 0
Precision becomes **0 or undefined**

✅ Shows:

Model is not even attempting positive predictions

Accuracy hides this — precision exposes it.

🎯 3 F1 Score

Formula:
[
F1 = harmonic\ mean\ of\ precision\ and\ recall
]

If either precision or recall is low → F1 becomes low.

Here:
Precision ≈ 0
Recall = 0

F1 = **0**

✅ F1 punishes models that ignore minority class
Accuracy does not.

🎯 4 Confusion Matrix

Accuracy gives **one number**
Confusion matrix shows **all four outcomes**

You immediately see:

	Pred +	**Pred -**
Real +	0	10 ❌
Real -	0	990

✅ You visually see:

All positives are missed

That's why confusion matrix is essential for imbalanced data.

🎯 5 ROC-AUC

ROC curve plots:

Recall (TPR) vs False Positive Rate

A model that predicts all negative:

- TPR = 0

- Curve hugs bottom
- AUC ≈ **0.5 (random)**

✅ Shows model has **no discrimination power**

🎯 6 PR-AUC (Even Better for Imbalance)

Precision-Recall curve focuses only on **positive class performance**

Why better than ROC in imbalance:

- ROC includes TN heavily (majority class dominates)
- PR curve ignores TN → focuses on minority detection

If model misses positives → PR-AUC drops sharply

✅ More sensitive to minority class failure

❓ Easy Memory Rule

Accuracy asks:

“How many total correct?”

Other metrics ask:

“How well did you detect the rare important class?”

🎓 Exam One-Line Answer

These metrics are better because they:

Evaluate performance specifically on the positive/minority class instead of being dominated by the majority class like accuracy.